1. Analyse Various Planning Approaches in Detail. [9]

Introduction

Planning in Artificial Intelligence is the process of finding a sequence of actions that helps an agent achieve a goal from an initial state.

A planning system decides:

- What actions to perform
- In what order to perform them
- How to achieve the desired goal efficiently

Different planning approaches are used depending on the problem environment.

Various Planning Approaches

1) Classical Planning

Classical planning is the basic form of AI planning where:

- Environment is completely known
- Actions are deterministic
- Only one agent is involved
- Initial state and goal state are clearly defined

The planner searches for a sequence of actions to move from initial state to goal state.

Features

- Fully observable environment
- Static world
- Deterministic actions
- Sequential execution

Example

Robot navigation:

- Initial state: Robot in Room A
- Goal state: Robot in Room B
- Action: Move(A,B)

Advantages

- Simple and easy to implement
- Gives optimal solutions

Disadvantages

- Not suitable for uncertain environments
- Cannot handle dynamic changes

2) Hierarchical Planning

Hierarchical planning divides a large problem into smaller sub-problems.

High-level goals are broken into smaller tasks until simple executable actions are obtained.

This method is also called:

Hierarchical Task Network (HTN) Planning

Working

1. Start with main task
2. Divide task into subtasks
3. Continue decomposition
4. Execute primitive actions

Example

Goal: Organize a seminar

Subtasks:

- Book hall
- Invite speakers
- Arrange food
- Prepare schedule

Advantages

- Reduces complexity
- Faster planning
- Easy management of large tasks

Disadvantages

- Requires domain knowledge
- Difficult to create hierarchy

3) Conditional Planning

Conditional planning is used when outcomes are uncertain.

The planner creates different plans for different conditions.

Features

- Handles uncertainty
- Uses “if-then” conditions
- Multiple possible outcomes

Example
Robot delivery:

- If road is clear → move ahead
- Else → choose alternate route

Advantages

- Flexible planning
- Works in uncertain environments

Disadvantages

- Planning becomes complex
- Requires more computation

4) Reactive Planning
Reactive planning responds immediately to environmental changes.
Instead of creating a complete plan in advance, the agent reacts to current situations.
Features

- Real-time decision making
- Fast response
- Suitable for dynamic environments

Example
Self-driving car reacting to traffic signals and pedestrians.
Advantages

- Quick response
- Suitable for dynamic systems

Disadvantages

- May not produce optimal plan
- Less future prediction

5) Probabilistic Planning
In probabilistic planning, actions may have uncertain results with certain probabilities.
Example
Action: Throw ball

- Hit target with probability 0.8
- Miss target with probability 0.2

Applications

- Robotics
- Medical diagnosis
- Weather forecasting

Advantages

- Handles uncertainty effectively
- Realistic planning

Disadvantages

- Complex calculations
- High computational cost

6) Temporal Planning
Temporal planning considers time constraints while planning actions.
The planner decides:

- When action should start
- Duration of action
- Scheduling of tasks

Example
Factory scheduling:

- Machine A works from 10 AM to 11 AM
- Packing starts after production ends

Advantages

- Efficient resource usage
- Handles deadlines

Disadvantages

- Complex scheduling
- Difficult optimization

7) Multi-Agent Planning
Multiple agents work together to achieve a common goal.
Agents coordinate and share tasks.
Example
Warehouse robots moving products together.
Advantages

- Parallel execution
- Faster completion

Disadvantages

- Coordination difficulty
- Communication overhead

Comparison of Planning Approaches

Planning Approach	Environment	Main Feature	Example
Classical Planning	Certain	Fixed sequence of actions	Robot navigation
Hierarchical Planning	Structured	Divide tasks into subtasks	Seminar organization
Conditional Planning	Uncertain	If-then plans	Alternate routes
Reactive Planning	Dynamic	Immediate response	Self-driving car
Probabilistic Planning	Uncertain	Probability-based actions	Weather prediction
Temporal Planning	Time-based	Scheduling and timing	Factory scheduling
Multi-Agent Planning	Cooperative	Multiple agents	Warehouse robots

---

2. Discuss AI and its Ethical Concerns, Explain Limitations of AI. [8]

Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that develops intelligent systems capable of performing tasks that normally require human intelligence.

These tasks include:

- Learning
- Decision making
- Language understanding
- Reasoning
- Problem solving

Examples:

- Chatbots
- Recommendation systems
- Self-driving cars
- Face recognition

Ethical Concerns of AI

AI systems can create many ethical and social problems if not used properly.

1) Privacy Issues - AI collects large amounts of personal data.

Problems

- Data misuse
- Unauthorized access
- Surveillance

Example - Social media platforms tracking user activities.

2) Bias and Discrimination - AI systems may become biased if training data is biased.

Problems

- Unfair decisions
- Discrimination in hiring or loans

Example - AI recruitment system favoring certain candidates.

3) Job Displacement - Automation can replace human workers.

Affected Areas

- Manufacturing
- Customer support
- Transportation

Result

- Unemployment
- Economic imbalance

4) Lack of Transparency

Many AI systems work like a “black box.”

People cannot understand how decisions are made.

Problem

- Difficult to trust AI decisions

5) Security Threats - AI can be used for harmful purposes.

Examples

- Cyber attacks
- Deepfakes
- Autonomous weapons

6) Dependency on Machines - Humans may become too dependent on AI systems.

Problems

- Reduced human skills
- Less critical thinking

7) Ethical Decision Making - AI may face situations involving moral decisions.

Example - Self-driving car accident decisions.

Limitations of AI - Although AI is powerful, it still has many limitations.

1) Lack of Human Intelligence - AI cannot think like humans.

It lacks:

- Common sense
- Emotions
- Creativity
- Moral values

2) High Cost - Developing AI systems requires:

- Expensive hardware
- Large datasets
- Skilled experts

3) Dependency on Data

AI performance depends heavily on training data.

Poor data leads to poor results.

4) No Emotional Understanding - AI cannot truly understand human emotions and feelings.

Example - Chatbots may fail to understand emotional situations.

5) Limited Creativity - AI can generate content but lacks true imagination and originality.

6) Requires Huge Computing Power

Advanced AI models need:

- High memory
- GPUs
- Large energy consumption

7) Cannot Handle Unexpected Situations Well

AI works best in trained conditions.

Unexpected situations may cause failure.

Example - Autonomous vehicles in unusual weather conditions.

8) Risk of Errors - Wrong predictions or biased outputs can lead to serious problems.

Example - Incorrect medical diagnosis.

3. Explain the Terms for Time and Schedule from Perspective of Temporal Planning. [9]

Introduction

Temporal planning is a planning method in AI that considers:

- Time
- Duration
- Scheduling
- Resource allocation

In temporal planning, actions are not instantaneous.

Each activity requires time and proper scheduling.

Time in Temporal Planning

Time represents when an action starts, ends, and how long it takes.

AI systems use time to organize activities efficiently.

Important Terms Related to Time

1) Time Point - A specific instant at which an event occurs.

Example

- Start production at 10:00 AM

2) Time Interval - The duration between two time points.

Example

- Meeting from 2 PM to 3 PM

3) Action Duration - The amount of time needed to complete an action.

Example

- Charging battery takes 2 hours

4) Temporal Constraints - Restrictions related to timing of actions.

Types

- Before constraint

- After constraint
- Simultaneous execution

Example - Cooking must finish before serving food.
5) Deadlines - Specific time before which a task must be completed.
Example - Project submission before 5 PM.
6) Concurrent Actions - Multiple actions executed at the same time.
Example - Printing and scanning simultaneously.
Scheduling in Temporal Planning
Scheduling is the process of assigning:

- Start time
- End time
- Resources
  to different tasks.

The objective is to complete all tasks efficiently.
Components of Scheduling
1) Tasks - Activities to be completed.
Example

- Manufacturing
- Packing
- Delivery

2) Resources - Resources required to perform tasks.
Types

- Machines
- Workers
- Memory
- Energy

3) Precedence Constraints - One task must finish before another starts.
Example - Compilation before execution.
4) Resource Constraints - Limited availability of resources.
Example - One machine cannot process two jobs simultaneously
5) Start Time and Finish Time
Every task has:

- Start time
- Completion time

Types of Scheduling
1) Static Scheduling - Schedule is fixed before execution.

Advantage	**Disadvantage**
Simple implementation	Cannot handle changes

2) Dynamic Scheduling - Schedule changes according to environment.

Advantage	**Disadvantage**
Flexible	Complex computation

Applications of Temporal Planning

- Robot task planning
- Factory automation
- Space missions
- Airline scheduling
- Project management

Example of Temporal Planning
Suppose a robot performs:

1. Pick object – 2 minutes
2. Move object – 3 minutes
3. Place object – 1 minute

Constraints:

- Move starts after pick
- Place starts after move

Scheduler arranges tasks according to time sequence.

---

4. Write a Detailed Note on AI Architecture. [8]
Introduction

AI Architecture is the structural design of an Artificial Intelligence system.
It defines:

- Components of AI system
- Data flow
- Decision-making process
- Interaction between modules

AI architecture helps in building intelligent systems capable of learning and reasoning.

Components of AI Architecture

1) Knowledge Base - Stores facts, rules, and information used by the AI system.

Functions

- Stores domain knowledge
- Supports reasoning
- Helps decision making

Example - Medical diagnosis database.

2) Learning Component - Allows the system to learn from data and improve performance.

Techniques

- Machine Learning
- Deep Learning
- Reinforcement Learning

Example - Spam email filtering.

3) Reasoning Engine - Performs logical thinking and decision making.

Functions

- Problem solving
- Inference generation
- Rule application

Example - Expert systems.

4) Perception Module - Collects information from environment using sensors or inputs.

Inputs

- Images
- Audio
- Text
- Sensor data

Example - Face recognition systems.

5) Planning Module - Generates sequence of actions to achieve goals.

Functions

- Goal formulation
- Action selection
- Scheduling

Example - Robot navigation planning.

6) Natural Language Processing (NLP) - Enables AI to understand human language.

Applications

- Chatbots
- Voice assistants
- Language translation

7) Execution Module - Executes selected actions in the environment.

Example- Robot arm movement.

Working of AI Architecture

1. Input is collected through sensors
2. Data is processed
3. Knowledge base is accessed
4. Reasoning engine makes decisions
5. Planning module selects actions
6. Execution module performs actions
7. Learning module improves future performance

Types of AI Architecture

1) Reactive Architecture - AI reacts directly to current inputs.

Features

- Fast response
- No memory usage

Example - Simple robots

2) Deliberative Architecture - AI uses reasoning and planning before action.

Features

- Goal-oriented
- Uses internal models

Example - Expert systems

3) Hybrid Architecture - Combination of reactive and deliberative systems.

Features

- Fast response
- Intelligent planning

Example - Self-driving cars

Diagram of AI Architecture

Sensors/Input
↓
Perception Module
↓
Knowledge Base
↓
Reasoning Engine
↓
Planning Module
↓
Execution Module
↓
Environment
↑
Learning Component

Applications of AI Architecture

- Robotics
- Autonomous vehicles
- Industrial automation
- Healthcare
- Smart assistants
- Banking systems

---

5. Explain with an Example State Space Planning. [5]

Introduction

State Space Planning is a planning technique in Artificial Intelligence where the planner searches through different states to find a path from the initial state to the goal state.

A state represents a condition or situation of the environment.

The planner applies actions to move from one state to another until the goal is achieved.

Components of State Space Planning

1) Initial State - The starting condition of the problem.

2) Goal State - The desired final condition.

3) Actions/Operators - Operations used to move from one state to another.

4) State Space - Collection of all possible states.

Types of State Space Planning

1) Forward Planning (Progression Planning)

- Starts from initial state
- Applies actions step by step
- Moves toward goal state

Flow

Initial State → Intermediate States → Goal State

2) Backward Planning (Regression Planning)

- Starts from goal state
- Works backward toward initial state
- Finds actions needed to achieve goal

Flow

Goal State → Previous States → Initial State

Example of State Space Planning

Problem - A robot is in Room A and wants to reach Room C.

States

- State 1: Robot in Room A
- State 2: Robot in Room B

- State 3: Robot in Room C

Actions

- Move(A,B)
- Move(B,C)

Forward Planning

Initial State:
Robot in A
Step 1:
Apply Move(A,B)
New State:
Robot in B
Step 2:
Apply Move(B,C)
Goal State Achieved:
Robot in C

Diagram

Room A ----> Room B ----> Room C
Initial Intermediate Goal

Advantages

- Simple and easy to understand
- Useful for path finding and robotics
- Systematic search process

Disadvantages

- Large state space increases complexity
- May require high memory and time

---

6. Explain with Example, How Planning is Different from Problem Solving. [5]

Introduction

Both planning and problem solving are important techniques in Artificial Intelligence.
They are used to achieve goals, but their approaches are different.
Problem solving mainly focuses on finding a solution path, while planning focuses on selecting proper actions and ordering them intelligently.

Difference Between Planning and Problem Solving

Planning	**Problem Solving**
Planning focuses on sequence of actions to achieve goals	Problem solving focuses on finding solution to a problem
Uses knowledge about actions and environment	Uses search techniques mainly
More suitable for complex real-world tasks	Suitable for well-defined problems
Handles time, resources and constraints	Usually ignores scheduling and resources
Generates optimized action plans	Generates path from initial state to goal
Used in robotics and automation	Used in puzzles and games

Example of Problem Solving

8-Puzzle Problem - The system searches different board states until the final arrangement is achieved.
Focus: Finding correct path to goal state.

Example of Planning

Robot Delivery System

Goal: Deliver package to another room.
Planner decides:

- Pick package
- Choose shortest path
- Avoid obstacles
- Deliver package

Focus: Action sequencing and execution.

Key Point

Problem solving answers:
“What is the solution?”
Planning answers:
“What actions should be performed and in what order?”

---

7. Explain AI Components and AI Architecture. [8]
Introduction
Artificial Intelligence systems are made of several components that work together to perform intelligent tasks.
AI architecture defines how these components are organized and connected.
AI Components
1) Learning Component - This component enables the AI system to learn from experience and improve performance.
Functions
- Data analysis
- Pattern recognition
- Model training

Example - Recommendation systems learning user preferences.
2) Knowledge Base - Stores information, facts, rules and relationships.
Functions
- Knowledge storage
- Information retrieval
- Decision support

Example - Medical expert systems.
3) Reasoning Component - Performs logical thinking and decision making.
Functions
- Drawing conclusions
- Solving problems
- Applying rules

Example - Chess-playing AI.
4) Perception Component - Collects input from environment.
Inputs
- Images
- Speech
- Sensors
- Text

Example - Face recognition system.
5) Planning Component - Creates sequence of actions to achieve goals.
Functions
- Goal formulation
- Action scheduling
- Path finding

Example - Robot navigation.
6) Natural Language Processing (NLP) - Allows machines to understand and process human language.
Applications
- Chatbots
- Voice assistants
- Translation systems

7) Execution Component - Executes actions decided by the AI system.
Example - Robot arm movement.
AI Architecture
AI Architecture is the structural design of an AI system that shows how components interact.
It controls:
- Information flow
- Learning process
- Decision making
- Action execution

Working of AI Architecture
1. Input is collected from environment
2. Perception module processes input

3. Knowledge base stores information
4. Reasoning engine analyzes data
5. Planning module creates action plan
6. Execution module performs actions
7. Learning module improves future performance

Types of AI Architecture

1) Reactive Architecture

- Responds directly to inputs
- No memory usage
- Fast response

Example - Simple robots

2) Deliberative Architecture

- Uses reasoning and planning
- Maintains internal model

Example - Expert systems

3) Hybrid Architecture - Combination of reactive and deliberative architecture.

Example - Self-driving cars

Diagram of AI Architecture

Sensors/Input
↓
Perception Module
↓
Knowledge Base
↓
Reasoning Engine
↓
Planning Module
↓
Execution Module
↓
Environment
↑
Learning Component

Applications

- Robotics
- Healthcare
- Smart assistants
- Autonomous vehicles
- Industrial automation

---

8. Explain Planning in Non-Deterministic Domain. [5]

Introduction

Planning in a non-deterministic domain refers to planning in environments where actions may produce different possible outcomes.

The result of an action is uncertain and cannot always be predicted exactly.

AI systems must prepare for multiple situations.

Non-Deterministic Domain

In deterministic environments:

- One action gives one predictable result.

In non-deterministic environments:

- One action may produce many possible results.

Example

Action:

Open Door

Possible outcomes:

- Door opens successfully

- Door remains locked
- Alarm activates

The AI system must create plans for all possibilities.

Features of Planning in Non-Deterministic Domain

1) Uncertain Outcomes - Actions may not always give same result.

2) Conditional Planning - Uses if-then conditions.

Example

If battery low → recharge

Else → continue task

3) Contingency Planning - Creates backup plans for unexpected situations.

4) Dynamic Environment - Environment changes during execution.

Types of Planning Used

1) Conditional Planning - Plan changes according to conditions.

2) Probabilistic Planning - Uses probabilities for outcomes.

3) Reactive Planning - Agent reacts immediately to changes.

Example of Robot Navigation

A robot moves toward destination.

Possible situations:

- Path clear
- Obstacle detected
- Battery low

Robot must choose different actions depending on situation.

Advantages

- Handles real-world uncertainty
- Flexible decision making
- Useful in robotics and automation

Disadvantages

- Complex planning process
- Requires high computation
- Difficult to predict all outcomes

---

9. Explain. [8]

i) Importance of Planning

Introduction

Planning is an important concept in Artificial Intelligence that helps an agent decide what actions should be taken to achieve a goal.

A planning system creates a sequence of actions before execution.

Planning is widely used in:

- Robotics
- Scheduling
- Game playing
- Automation
- Navigation systems

Importance of Planning in AI

1) Goal Achievement

Planning helps AI systems achieve desired goals systematically.

The planner selects proper actions to move from initial state to goal state.

Example - Robot planning shortest path to destination.

2) Better Decision Making

Planning helps AI choose the best possible action among many alternatives.

It improves intelligent decision making.

3) Reduces Complexity

Complex problems are divided into smaller manageable tasks.

This makes problem solving easier.

Example

Travel planning:

- Book tickets
- Reserve hotel
- Arrange transport

4) Efficient Resource Utilization

Planning helps use:

- Time
- Energy
- Memory
- Machines efficiently.

Example - Factory scheduling systems.

5) Saves Time and Cost

Proper planning avoids unnecessary actions and reduces wastage.

This improves system efficiency.

6) Handles Dynamic Environments

AI planning helps systems react to changing environments.

Example - Self-driving cars avoiding traffic.

7) Supports Automation

Planning is essential for automated systems.

Applications

- Industrial robots
- Smart homes
- Automated warehouses

8) Improves Accuracy

Planned actions reduce errors and increase reliability.

Applications of Planning

- Robotics
- Space missions
- Medical diagnosis
- Logistics
- Autonomous vehicles
- Project management

ii) Algorithm for Classical Planning

Introduction - Classical planning is a planning approach where:

- Environment is fully observable
- Actions are deterministic
- Initial and goal states are known

The planner searches for a sequence of actions to achieve the goal.

Classical Planning Algorithm

The algorithm works by exploring states and applying actions until the goal state is reached.

Steps of Classical Planning Algorithm

Step 1: Define Initial State - Specify the starting condition of the problem.

Example - Robot is in Room A.

Step 2: Define Goal State - Specify desired final condition.

Example - Robot should reach Room C.

Step 3: Define Actions/Operators - Specify possible actions and their effects.

Example

- Move(A,B)
- Move(B,C)

Each action contains:

- Preconditions
- Effects

Step 4: Check Applicable Actions - Planner checks which actions can be applied in current state.

Step 5: Generate New States - Apply action and move to next state.

Step 6: Goal Test - Check whether current state matches goal state.

- If yes → stop
- Else → continue search

Algorithm

1. Start with initial state
2. Compare current state with goal state
3. If goal reached, stop
4. Else find applicable actions
5. Apply action to generate new state
6. Repeat process until goal is achieved

Example

Initial State:

Robot in A

Goal State:

Robot in C

Actions:

- Move(A,B)
- Move(B,C)

Execution:

- Apply Move(A,B)
- Apply Move(B,C)

Goal achieved.

Advantages

- Simple and systematic
- Produces valid plans
- Useful for structured environments

Disadvantages

- Not suitable for uncertain environments
- Large state spaces increase complexity

---

10. Explain Limits of AI and Future Opportunities with AI. [5]

Limits of AI

Although Artificial Intelligence is powerful, it has several limitations.

1) Lack of Human Emotions

AI cannot truly understand feelings, emotions or empathy.

2) No Common Sense

AI works only on trained data and rules.
It cannot think like humans in unexpected situations.

3) Dependency on Data

AI requires huge amounts of quality data.
Poor data leads to poor results.

4) High Cost

AI systems require:

- Expensive hardware
- Skilled professionals
- Large computational resources

5) Risk of Errors

Wrong predictions may cause serious problems.

Example

Medical diagnosis errors.

6) Limited Creativity

AI can generate content but lacks real imagination and originality.

Future Opportunities with AI

AI has a very bright future in many fields.

1) Healthcare - AI can help in:

- Disease prediction
- Medical imaging
- Robot surgery

2) Education - AI-based smart learning systems and virtual tutors can improve education.

3) Agriculture - AI helps in:

- Crop monitoring
- Weather prediction
- Smart irrigation

4) Transportation - AI enables:

- Self-driving vehicles
- Traffic management
- Route optimization

5) Industrial Automation - AI-powered robots improve productivity and reduce manual work.

6) Cyber Security - AI can detect cyber attacks and suspicious activities.

7) Smart Cities - AI supports:

- Smart traffic systems
- Energy management
- Waste management

---

11. What is AI? Explain Scope of AI in All Walks of Life Also Explain Future Opportunities with AI. [8]

Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that develops systems capable of performing tasks that normally require human intelligence.

These tasks include:
- Learning
- Reasoning
- Problem solving
- Decision making
- Language understanding

Scope of AI in Different Fields - AI is used in almost every area of life.

---

1) Healthcare - AI applications:
- Disease diagnosis
- Medical image analysis
- Robot-assisted surgery
- Virtual health assistants

Example - AI detecting cancer from X-ray images.
2) Education - AI improves personalized learning.
Applications
- Smart tutors
- Online learning systems
- Automatic grading

3) Banking and Finance - AI helps in:
- Fraud detection
- Online banking
- Credit scoring
- Stock market prediction

4) Agriculture - AI is used for:
- Crop monitoring
- Soil analysis
- Pest detection
- Smart irrigation

5) Transportation - AI applications:
- Self-driving cars
- Traffic prediction
- Navigation systems

6) Manufacturing Industry - AI-powered robots perform:
- Assembly
- Quality testing
- Packaging

7) E-Commerce - AI provides:
- Product recommendations
- Chatbots
- Customer support

Example - Online shopping recommendations.
8) Entertainment - AI is used in:
- Video recommendations
- Gaming
- Music suggestions

9) Security and Defense - AI supports:
- Surveillance systems
- Face recognition
- Cyber security

10) Smart Homes - AI controls:
- Smart lights
- Voice assistants
- Security systems

Future Opportunities with AI
1) Advanced Healthcare Systems - AI may help in early disease detection and personalized medicine.
2) Fully Autonomous Vehicles - Future vehicles may operate without human drivers.
3) Smart Robotics - Intelligent robots may assist humans in industries and homes.
4) AI in Space Research - AI can support:
- Space exploration
- Satellite monitoring
- Mission planning

5) Intelligent Education Systems - Future AI tutors may provide personalized education to every student.
6) Better Cyber Security - AI systems will improve protection against cyber attacks.
7) Human-AI Collaboration - Humans and AI will work together to improve productivity and innovation.

---

12. Explain with an Example Goal Stack Planning (STRIPS Algorithm). [5]
Introduction
Goal Stack Planning is a planning method used in Artificial Intelligence to achieve goals step by step.
It uses a stack data structure to store:
- Goals
- Subgoals
- Actions

STRIPS stands for: **Stanford Research Institute Problem Solver**
It is one of the most popular planning systems in AI.

Working of Goal Stack Planning

The planner:

1. Pushes goal onto stack
2. Checks whether goal is satisfied
3. If not satisfied, selects action
4. Pushes action preconditions as subgoals
5. Executes actions when preconditions are satisfied

Components of STRIPS

1) Initial State - Starting condition.

2) Goal State - Desired final condition.

3) Operators/Actions - Actions used to move between states.

Each action contains:

- Preconditions
- Add list
- Delete list

Example

Problem - A robot wants to move a box from Room A to Room B.

Initial State

- Robot in Room A
- Box in Room A

Goal State

- Box in Room B

Action

Move_Box(A,B)

Preconditions

- Robot in Room A
- Box in Room A

Add List

- Box in Room B

Delete List

- Box in Room A

Goal Stack Process

Step 1

Push goal:

- Box in Room B

Step 2

Select action:

- Move_Box(A,B)

Push preconditions:

- Robot in Room A
- Box in Room A

Step 3

Preconditions satisfied.

Execute action:

- Move_Box(A,B)

Step 4

Goal achieved:

- Box in Room B

Diagram

Goal Stack

Top

```
|
| Box in Room B
| Move_Box(A,B)
```

| Robot in Room A
| Box in Room A
|
Bottom

Advantages

- Simple planning technique
- Efficient goal decomposition
- Easy implementation

Disadvantages

- Difficult for complex problems
- May fail in uncertain environments

---

13. Write a Short Note on Planning Agent, State Goal and Action Representation. [8]

Introduction

Planning in Artificial Intelligence involves generating a sequence of actions to achieve a goal.
A planning system mainly depends on:

- Planning Agent
- State Representation
- Goal Representation
- Action Representation

These elements help the AI system understand the environment and perform intelligent actions.

1) Planning Agent

A Planning Agent is an intelligent system that decides what actions should be performed to achieve a goal.
The agent:

- Observes environment
- Selects actions
- Creates plans
- Executes actions

Functions of Planning Agent

1) Goal Formulation - Determines what needs to be achieved.
2) Problem Formulation - Represents the problem in structured form.
3) Planning - Finds sequence of actions.
4) Execution - Performs selected actions.
5) Monitoring - Checks whether goal is achieved.

Example

Robot vacuum cleaner:

- Detects dirty area
- Plans cleaning path
- Moves and cleans automatically

2) State Representation

A state represents the condition of the environment at a particular time.
State representation describes:

- Objects
- Properties
- Relationships

It helps AI understand current situation.

Types of State Representation

1) Complete State Representation - Contains all information about environment.
2) Partial State Representation - Contains only important information.

Example

Robot navigation:
State:

- Robot at Room A
- Door is open

This information describes current environment.

Importance of State Representation

- Helps in decision making
- Reduces complexity
- Supports planning and search

3) Goal Representation

Goal representation defines the desired final state that the agent wants to achieve.

Goals guide the planning process.
Example
Initial State:
Robot in Room A
Goal State:
Robot in Room C
The planner tries to achieve the goal state.
Types of Goals
1) Single Goal - Only one objective.
2) Multiple Goals - Several objectives together.
Importance of Goal Representation

- Directs planning process
- Helps evaluate progress
- Improves efficiency

4) Action Representation
Actions are operations that change one state into another.
Action representation describes:

- Preconditions
- Effects
- Changes after execution

Components of Action Representation
1) Preconditions - Conditions required before action execution.
2) Add Effects - Conditions added after action.
3) Delete Effects - Conditions removed after action.
Example
Action:
Move(A,B)
Preconditions

- Robot in A
- Path available

Add Effect

- Robot in B

Delete Effect

- Robot in A

Diagram
Initial State → Action → New State
Applications

- Robotics
- Game AI
- Automated scheduling
- Navigation systems

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14. Explain Different Components of Planning System. [7]
Introduction
A Planning System in Artificial Intelligence generates a sequence of actions to achieve a desired goal.
The planning system contains several important components that work together for intelligent decision making.
Components of Planning System
1) Initial State - The initial state represents the starting condition of the environment.
It describes:

- Current objects
- Positions
- Conditions

Example - Robot is in Room A.
2) Goal State

Goal state represents the desired final condition.
The planner tries to transform initial state into goal state.
Example - Robot should reach Room C.
3) Actions/Operators - Actions are operations that move the system from one state to another.
Each action contains:
- Preconditions
- Effects

Example - Move(A,B)
4) State Space
State space is the collection of all possible states.
The planner searches through state space to find solution path.
5) Planning Algorithm - Planning algorithm generates action sequence to achieve goal.
Examples
- STRIPS
- Goal Stack Planning
- Partial Order Planning

6) Knowledge Base
Stores facts, rules and domain information.
It helps planner make intelligent decisions.
Example - Map information for navigation system.
7) Constraints - Constraints are limitations that affect planning.
Types
- Time constraints
- Resource constraints
- Ordering constraints

Example - Task must finish before deadline.
8) Execution System - Executes planned actions in real environment.
Example - Robot performing planned movement.
Diagram of Planning System
Initial State → Planning Algorithm → Action Sequence → Goal State
↑
Knowledge Base
Applications of Planning Systems
- Robotics
- Automated manufacturing
- Route planning
- Space missions
- Project scheduling

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15. Write Note on Hierarchical Task Network Planning. [5]
Introduction
Hierarchical Task Network (HTN) Planning is a planning technique in Artificial Intelligence where a large task is divided into smaller subtasks.
Complex problems are solved step by step using task decomposition.
Working of HTN Planning
The planner:
1. Starts with high-level task
2. Divides task into smaller subtasks
3. Continues decomposition
4. Executes simple primitive actions

Components of HTN Planning
1) Tasks - Activities to be performed.
Types
- Primitive Tasks
- Compound Tasks

2) Methods - Rules used to decompose compound tasks into subtasks.
3) Operators - Primitive executable actions.
Example
Goal
Organize College Seminar

Subtasks

- Book hall
- Invite guests
- Arrange food
- Prepare presentation

Each subtask can be further divided.

Diagram

Organize Seminar
  ↓
┌──────────────┐
↓ ↓ ↓
Book Invite Arrange
Hall Guests Food

Advantages

- Reduces problem complexity
- Faster planning
- Suitable for large tasks

Disadvantages

- Requires domain knowledge
- Difficult hierarchy design

Applications

- Robotics
- Project management
- Military planning
- Workflow automation

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16. Explain Classical Planning and its Advantages with Example. [6]

Introduction

Classical Planning is a planning method in AI where:

- Environment is fully observable
- Actions are deterministic
- Initial state and goal state are clearly known

The planner generates a sequence of actions to achieve goal state.

Features of Classical Planning

- Single agent environment
- Static world
- Deterministic actions
- Sequential execution

Components

1) Initial State - Starting condition.
2) Goal State - Desired final condition.
3) Actions - Operations used to move between states.

Example

Robot Navigation

Initial State: Robot in Room A
Goal State: Robot in Room C
Actions:

- Move(A,B)
- Move(B,C)

Working

Step 1: Robot moves from A to B
Step 2: Robot moves from B to C
Goal achieved.

Diagram

Room A → Room B → Room C

Advantages of Classical Planning

1) Simple and Easy - Easy to understand and implement.
2) Systematic Search - Uses organized search process.
3) Produces Valid Plans - Generates proper action sequence.

4) Useful for Structured Problems - Works efficiently in predictable environments.
5) Supports Automation - Useful in robotics and navigation systems.
Disadvantages

- Cannot handle uncertainty
- Difficult for dynamic environments
- Large state space problem

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17. What are the Types of Planning? Explain in Detail. [6]
Introduction
Planning in Artificial Intelligence is the process of selecting actions to achieve goals.
Different planning methods are used depending on environment and problem complexity.
Types of Planning
1) Classical Planning - Classical planning assumes:

- Fully observable environment
- Deterministic actions
- Known initial and goal states

Example - Robot path finding.
Advantage - Simple implementation.
2) Hierarchical Planning - Large tasks are divided into smaller subtasks.
Example - Organizing an event.
Advantage - Reduces complexity.
3) Conditional Planning - Creates different plans for different conditions.
Example - If traffic high → choose alternate route.
Advantage - Handles uncertainty.
4) Reactive Planning - Agent reacts immediately to environmental changes.
Example - Self-driving cars.
Advantage - Fast response.
5) Probabilistic Planning - Uses probability to handle uncertain outcomes.
Example - Weather forecasting.
Advantage - More realistic planning.
6) Temporal Planning - Considers time and scheduling constraints.
Example - Factory scheduling.
Advantage - Efficient time management.
7) Multi-Agent Planning - Multiple agents cooperate to achieve goals.
Example - Warehouse robots.
Advantage - Parallel task execution.
Comparison Table

Planning Type	Main Feature	Example
Classical Planning	Deterministic environment	Robot navigation
Hierarchical Planning	Task decomposition	Event organization
Conditional Planning	If-then planning	Route selection
Reactive Planning	Immediate response	Self-driving car
Probabilistic Planning	Probability-based actions	Weather prediction
Temporal Planning	Time constraints	Factory scheduling
Multi-Agent Planning	Multiple cooperating agents	Warehouse robots